EasyVisa Project

Post Graduate Program in Artificial Intelligence and Machine Learning

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February 21, 2026

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Executive Summary

The EasyVisa Machine Learning project was designed to predict whether a work visa application will be certified or denied prior to submission. The primary objective is to reduce approval risk, minimize unexpected denials, and provide smarter, data-driven guidance to clients.

During data analysis, several important patterns were identified. Higher education levels and prior job experience were associated with higher certification rates. Full-time roles were more likely to be approved. Prevailing wage levels influenced outcomes, and certain regions and employer characteristics played a meaningful role in certification decisions.

After evaluating multiple ensemble models, the **tuned XGBoost model** was selected due to its strong performance, achieving 94.7% recall and an 81.5% F1-score. Its high recall directly aligns with the business objective of minimizing high-risk visa approvals while maintaining balanced predictive performance.

Overall, this solution enables EasyVisa to proactively assess application risk, improve decision-making, optimize operational resources, and enhance client success rates.

Business Problem Overview and Solution Approach

Problem Overview:

EasyVisa provides end-to-end support to organizations and international talent navigating the U.S. work visa process. Visa outcomes are influenced by a wide range of variables, including candidate qualifications, wage compliance, employer characteristics, and regional labor market conditions, making the process both data-intensive and highly regulated. As a result, high-risk applications may be submitted without proactive risk mitigation, increasing the likelihood of unexpected denials for clients.

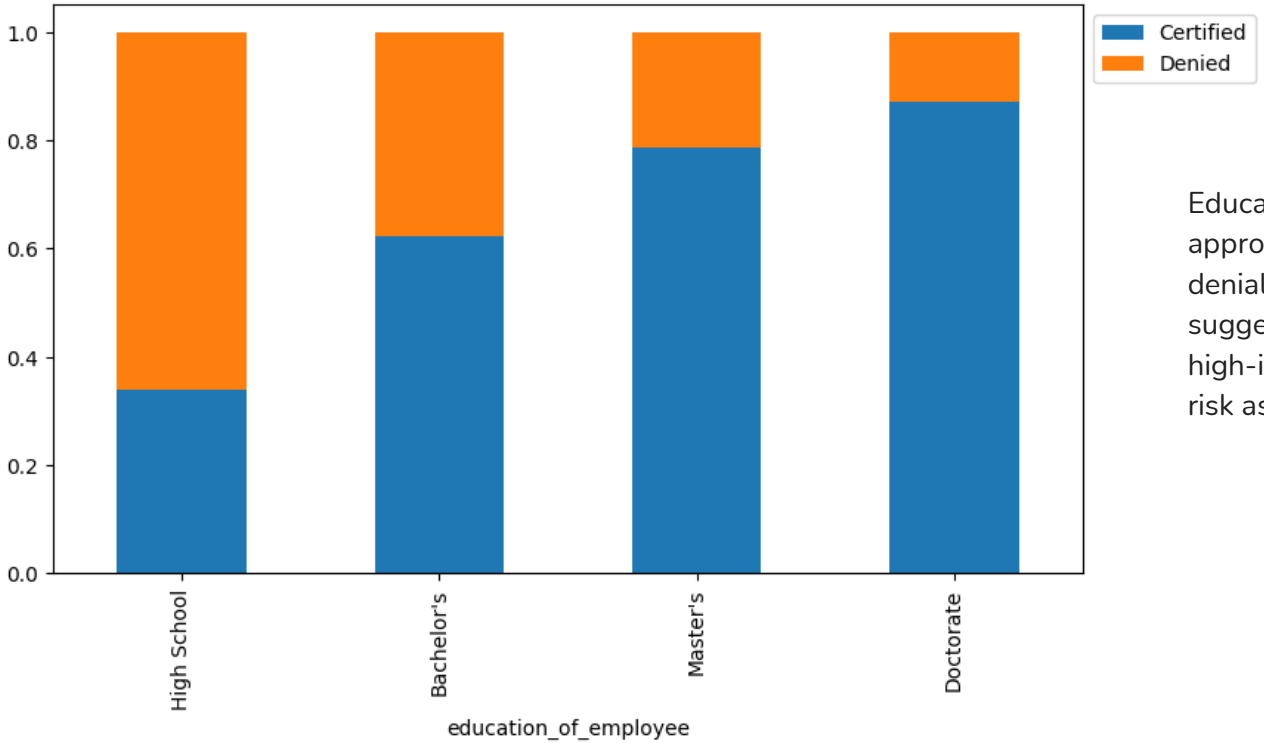
Solution Approach:

To address the problem, EasyVisa will develop a machine learning classification model that predicts whether a visa application is likely to be approved or denied based on historical data. The approach includes analyzing and preparing the data, building and comparing multiple predictive models, and selecting the best-performing model using evaluation metrics such as recall, precision, and F1-score. The final solution will provide actionable insights into key approval drivers and enable proactive risk assessment, helping improve decision-making, increase approval rates, and optimize operational efficiency.

EDA Observations

Does education have any impact on visa certification outcomes?

Education Level / Case Status

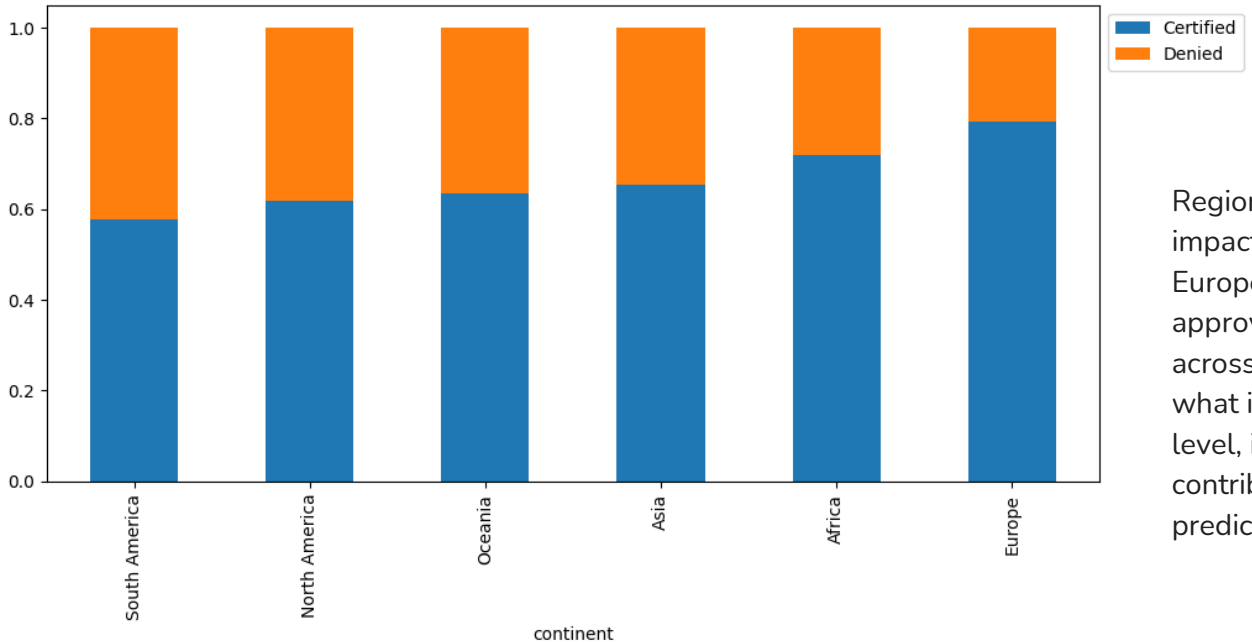


Education level is a major predictor of visa approval. As educational attainment increases, denial risk decreases significantly. This suggests education should be treated as a high-impact feature in predictive modeling and risk assessment for visa applications.

EDA Observations

How the visa status vary across different continents?

Region / Case Status

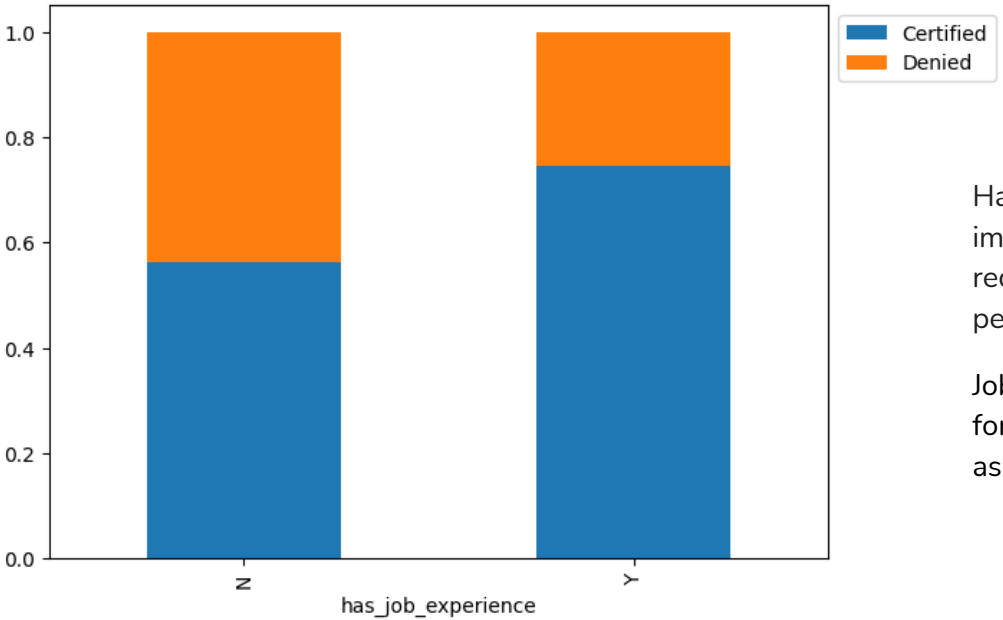


Region of origin appears to have a moderate impact on certification outcomes, with European applicants experiencing the highest approval likelihood. However, the variation across continents is less pronounced than what is typically observed with education level, indicating that continent may be a contributing factor but not the strongest predictor of approval.

EDA Observations

Does having prior work experience influence the likelihood of visa certification?

Job Experience / Case Status



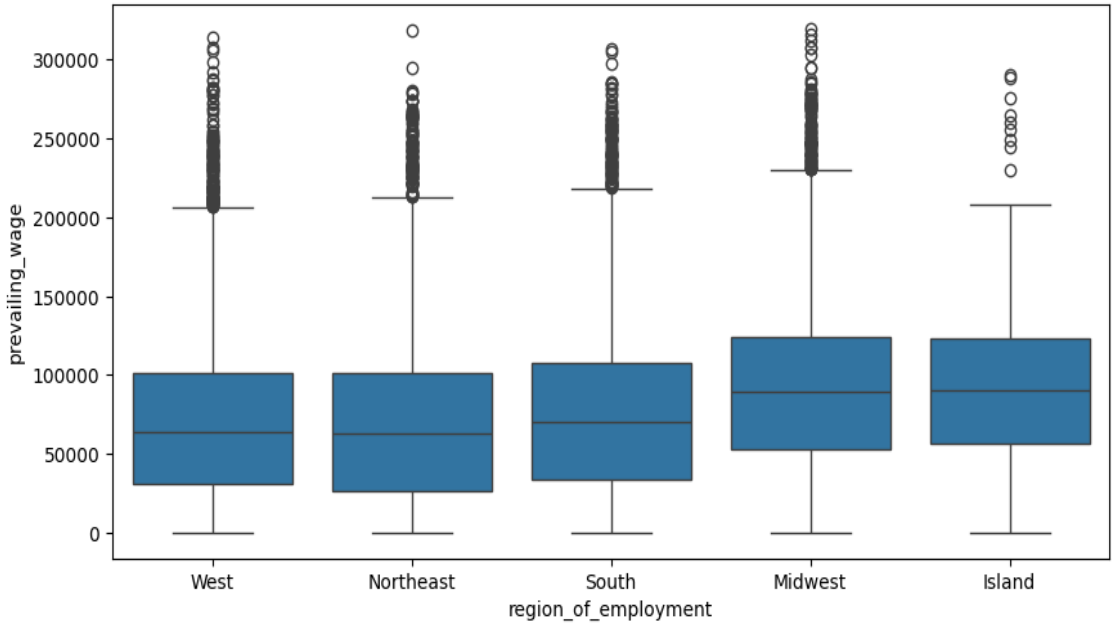
Having prior job experience materially improves the probability of certification and reduces denial risk. From a modeling perspective

Job experience is likely an important feature for predicting application outcomes and assessing approval probability

EDA Observations

Is the prevailing wage similar across all the regions of the US?

Region of Employment /Prevailing Wage

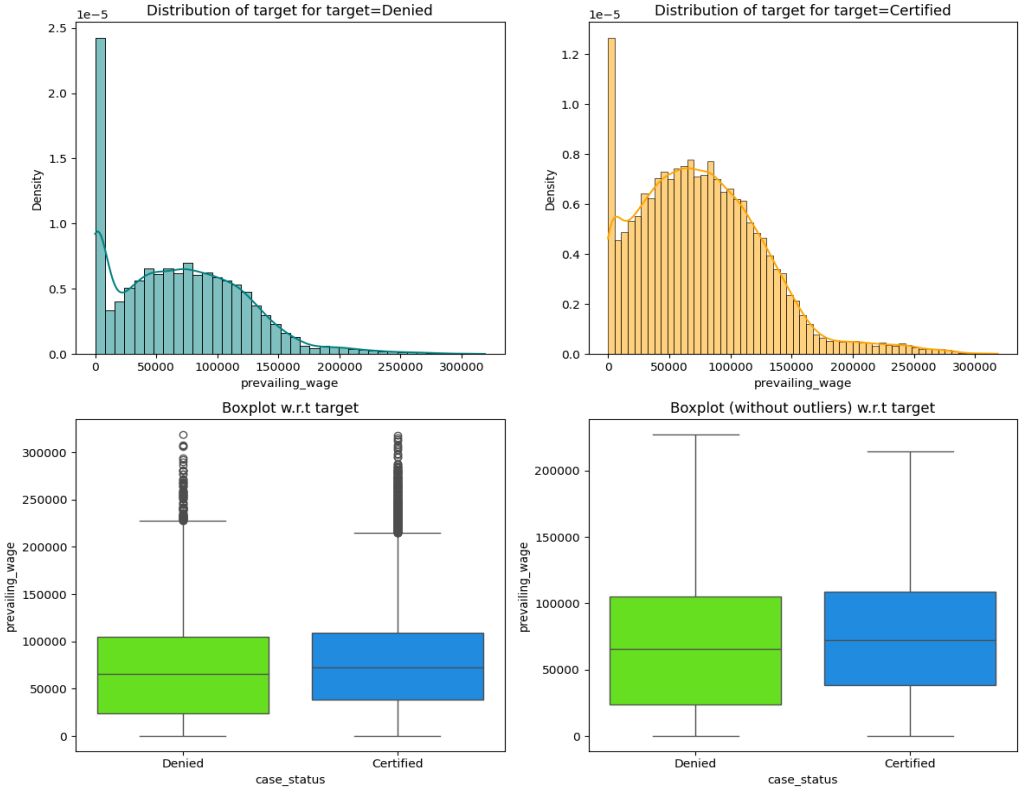


The prevailing wage varies across regions, with the **Midwest and Island** showing the highest median wages, followed by the **South**, while the **West and Northeast** have relatively lower median values. All regions display a wide spread of wages and significant high-end outliers, indicating the presence of some very high-paying roles

EDA Observations

Does the visa status change based on the prevailing wage?

Prevailing Wage / Case Status



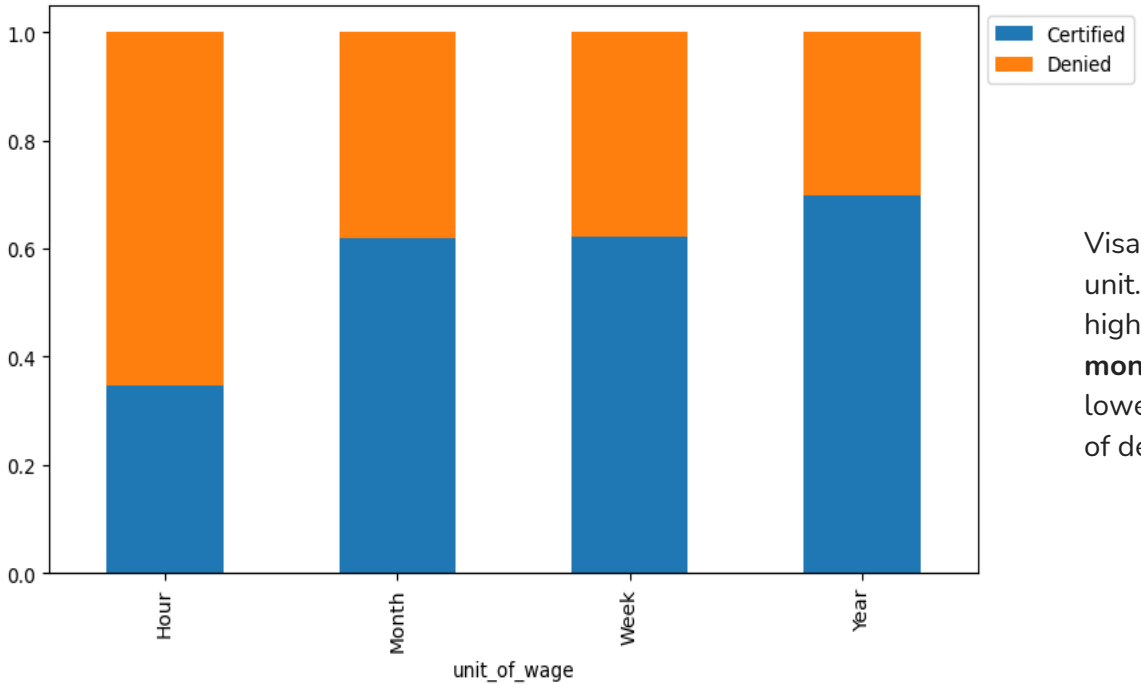
Certified visa applications tend to have higher prevailing wages compared to denied cases. The distribution for certified cases is slightly shifted toward higher salary ranges, and this difference remains even after removing outliers.

Both groups show right-skewed distributions with some very high-wage outliers, but the overall central tendency (median) is higher for certified applications.

EDA Observations

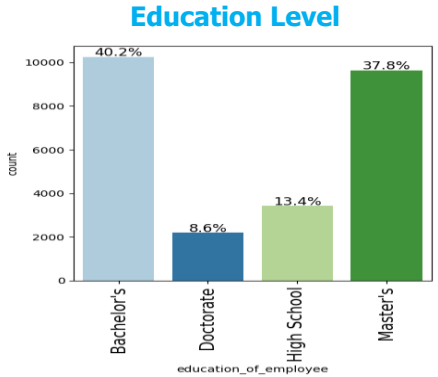
Does unit of wage have any impact on visa applications getting certified?

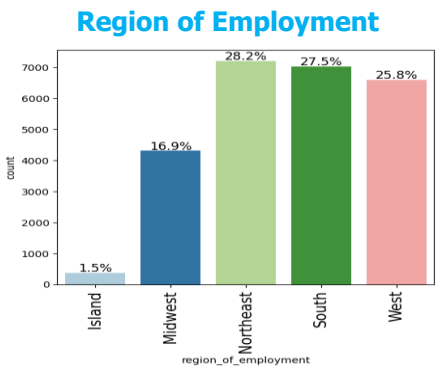
Unit of Wage / Case Status

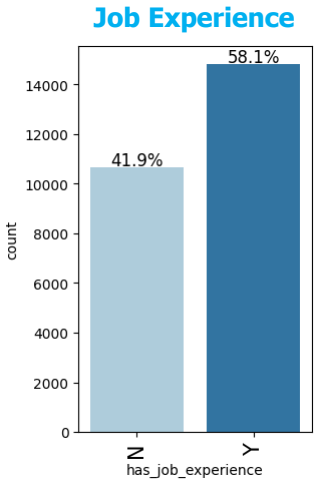


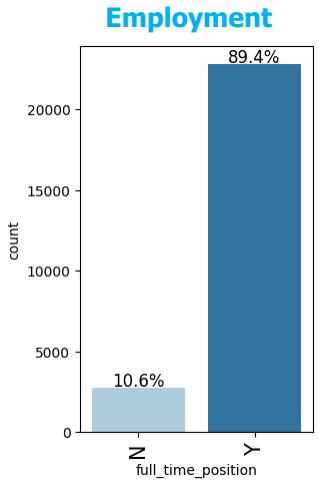
Visa certification rates vary significantly by wage unit. Applications with **yearly wages** have the highest approval rate, followed by **weekly and monthly wages**, while **hourly wages** have the lowest certification rate and the highest proportion of denials.

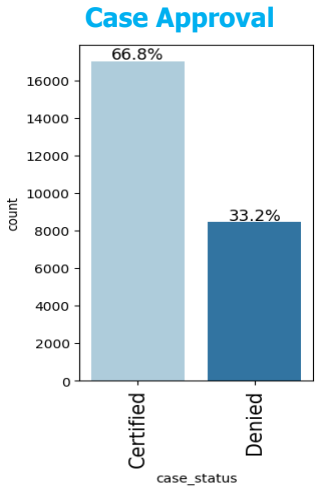
EDA Observations











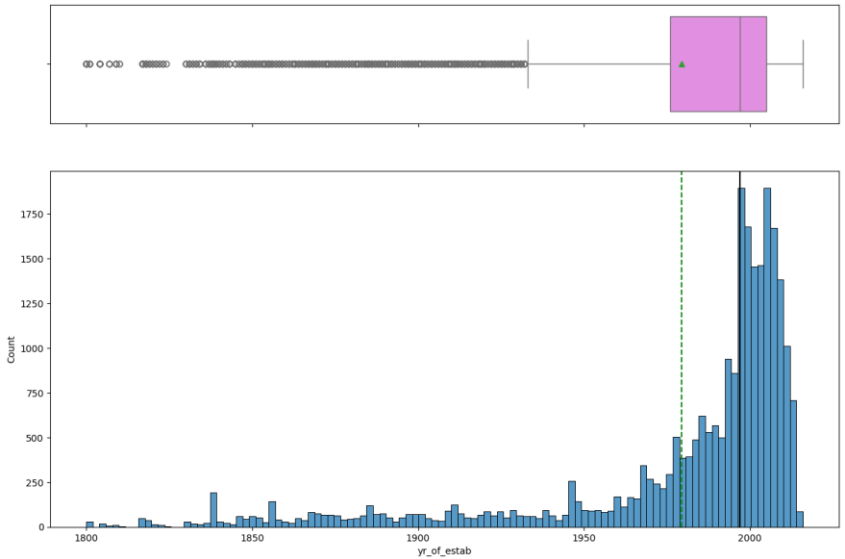
Education: Nearly 78% of applicants hold at least a Bachelor's degree. he dataset is highly skewed toward higher education levels.

Employment: Applications are fairly evenly distributed among Northeast, South, and West - Employment demand for foreign workers appears strongest in Northeast, South and West

Approximately **58% of applicants have prior job experience**, and nearly **90% of positions are full-time**, reflecting the structured nature of employment-based visa sponsorship. In terms of outcomes, about **67% of cases are certified**, while **33% are denied**, creating a moderate class imbalance in the target variable.

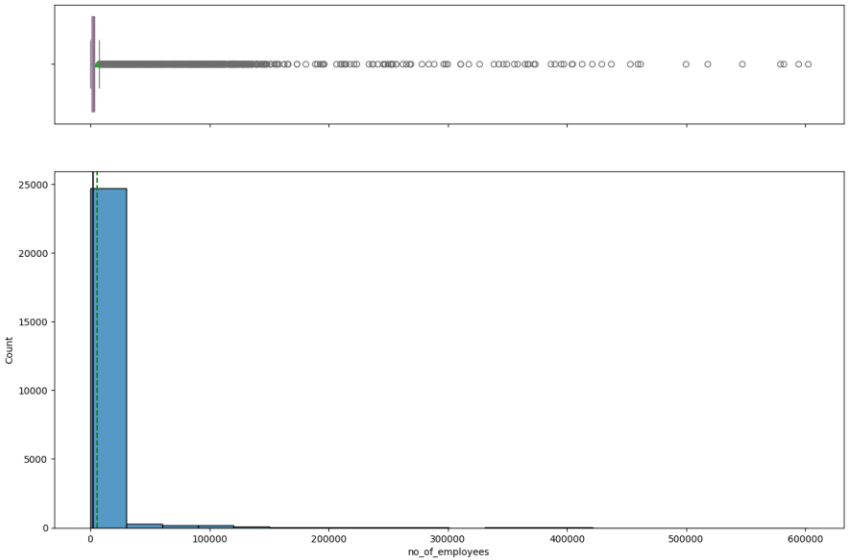
EDA Observations

Company Years of Establishment



- The distribution is heavily left-skewed
- Most companies were established **after 1970**, with a strong concentration between **1990–2015**.
- The median appears to be in the **late 1990s or early 2000s**, indicating most sponsoring companies are relatively modern.

Number of Employees



- The vast majority of companies have **relatively small employee counts**.
- A small number of companies have **very large employee sizes (100K–600K+)**, creating strong outliers.
- The median is much lower than the mean, confirming heavy skewness.

EDA Observations

Correlation Heat Map

There is **no significant linear correlation** among the numerical features, meaning each variable likely contributes independent information to the predictive model.

Data Preprocessing: Data Types

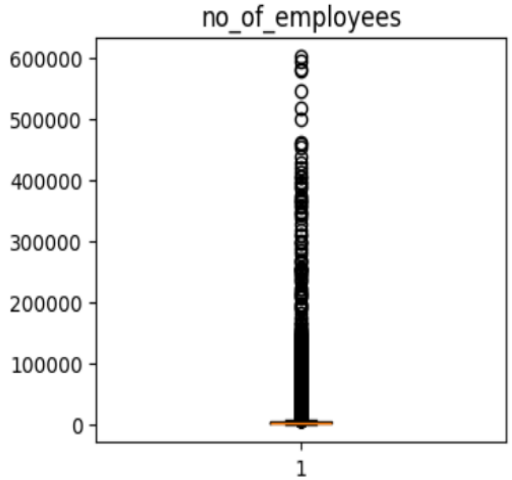
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Data Type

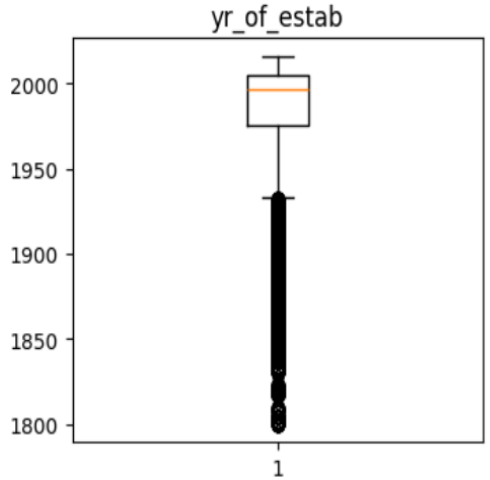
- Int 64: 2 columns (numerical)
- Float 64: 1 column (numerical)
- Object: 9 columns (numerical)
- Columns: 12
- Rows: 25480
- No missing values
- 'case_id' should be dropped
- Target variable: 'case_status'

Data Preprocessing: Outlier Detection

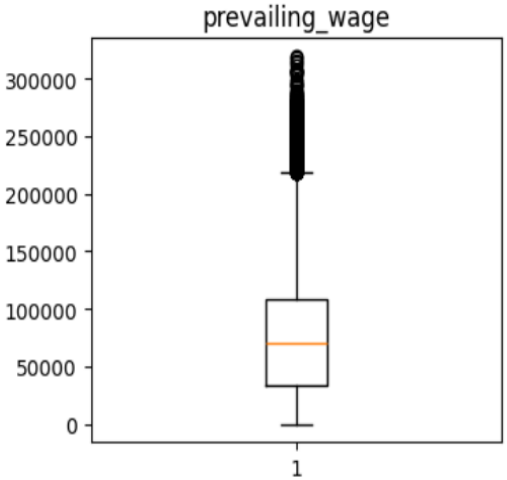
Outlier detection using Boxplot



- Large number of outliers
- Extremely right-skewed distribution
- High Variability



- Left Skewness
- Several lower end outliers
- High Variability



- Extremely right-skewed distribution
- High Variability

Data Preprocessing: Duplicate / Missing Values

Duplicate Values

Checking for duplicate values

```
# checking for duplicate values
data.duplicated().sum() ##  Complete the code to check for duplicate values
```

```
np.int64(0)
```

There are no duplicate values

Missing Values

Checking for missing values

```
data.isnull().sum() ##  Complete the code to check for missing values
```

	0
case_id	0
continent	0
education_of_employee	0
has_job_experience	0
requires_job_training	0
no_of_employees	0
yr_of_estab	0
region_of_employment	0
prevailing_wage	0
unit_of_wage	0
full_time_position	0
case_status	0

dtype: int64

There are no missing values

Data Preprocessing: Feature Engineering

Encoding categorical features

```
data["case_status"] = data["case_status"].apply(lambda x: 1 if x == "Certified" else 0)

X = data.drop(["case_status"], axis=1)
y = data["case_status"]


X = pd.get_dummies(X, drop_first=True)
# X = X.astype(float)
```

Target variable 'case_status' was encoded to numerical values to allow for the ML model to process the information:

- Certified: 1
- Denied: 0

Two Step Data Split

```
# Splitting data into training, validation and test set:
# first we split data into 2 parts, say temporary and test

X_temp, X_test, y_temp, y_test = train_test_split(
    X, y, test_size=0.2, random_state=1, stratify=y
)

# then we split the temporary set into train and validation

X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp, test_size=0.25, random_state=1, stratify=y_temp
)
```

The data set was split into 20% test and 80% training

The train data set was then split into 20% validation and 60% left for training

Validation data is required to help tune the model without touching the test data thus preventing data leaks.

- **Training Set: (15288, 21)**
- **Validation Set: (5096, 21)**
- **Test Set: (5096,21)**

Model Performance Strategy and Comparison

Data	Training	Validation	Sampling Strategy	Training Sample case_status
Original	• Bagging: 0.989150179193873 • Random Forest: 1.0 • **Gradient Boosting: 0.8291218182658106** • Adaboost: 0.8204269947530306 • XGBoost: 0.8963229453814886	• Bagging: 0.7736516357206012 • Random Forest: 0.8049978941457251 • **Gradient Boosting: 0.826637008202419** • Adaboost: 0.8180081855388813 • XGBoost: 0.8079119654547987	N/A	• Certified: 10210 • Denied: 5078
Oversampled	• Bagging: 0.9875473741201949 • Random Forest: 0.9999510260051913 • Gradient Boosting: 0.8072434234901815 • Adaboost: 0.8005498403689252 • XGBoost: 0.8708686342053813	• Bagging: 0.7665171898355755 • Random Forest: 0.7965442764578834 • Gradient Boosting: 0.8173049645390071 • Adaboost: 0.8195334879279771 • XGBoost: 0.8129304286718201	SMOTE Artificially increases the minority class samples : 'denied'	• Certified: 10210 • Denied: 10210
Undersampled	• Bagging: 0.9803687095166915 • Random Forest: 1.0 • Gradient Boosting: 0.7281441717791411 • Adaboost: 0.7015343047380103 • XGBoost: 0.8720351390922401	• Bagging: 0.7057046979865772 • Random Forest: 0.7417218543046358 • Gradient Boosting: 0.776595744680851 • Adaboost: 0.765990884802766 • XGBoost: 0.7459304181295883	Random Under Sample Decreases the majority class samples : 'certified'	• Certified: 5078 • Denied: 5078

This comparison evaluates five ensemble models (Bagging, Random Forest, Gradient Boosting, AdaBoost, XGBoost) across three data strategies: **Original, Oversampled, and Undersampled**, focusing on training vs validation performance.

Gradient Boosting on Original Data appears to be the strongest overall choice.

Hyperparameter Tuning: Random Forest (rf_tuned)

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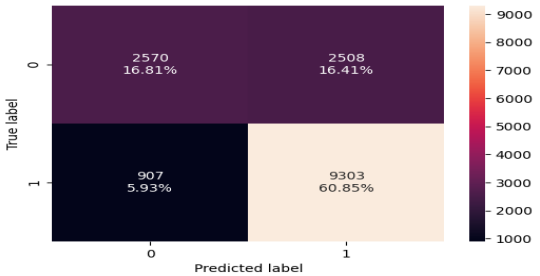
Tuning

Confusion Matrix

Performance Metrics

Training

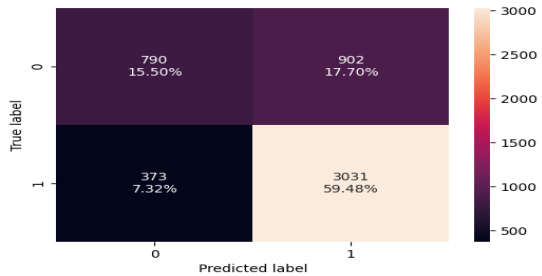
```
parameters = {
    "max_depth": list(np.arange(5, 15, 5)),
    "max_features": ["sqrt", "log2"],
    "min_samples_split": [3, 5, 7],
    "n_estimators": np.arange(10, 40, 10),
```



- **Accuracy: 0.77**
- **Recall: 0.91**
- **Precision: 0.78**
- **F1: 0.84**

Validation

```
parameters = {
    "max_depth": list(np.arange(5, 15, 5)),
    "max_features": ["sqrt", "log2"],
    "min_samples_split": [3, 5, 7],
    "n_estimators": np.arange(10, 40, 10),
```



- **Accuracy: 0.74**
- **Recall: 0.89**
- **Precision: 0.77**
- **F1: 0.82**

The tuned Random Forest model demonstrates strong validation performance, achieving high recall (89%) and a balanced F1-score (82%). This indicates effective detection of the target class while maintaining reasonable precision, suggesting good generalization.

Hyperparameter Tuning: AdaBoost (abc_tuned)

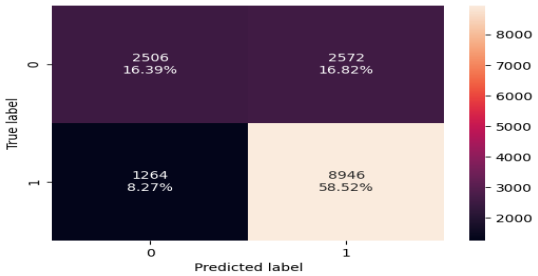
Tuning

Confusion Matrix

Performance Metrics

Training

```
parameters = {
    # Let's try different max_depth for base_estimator
    "estimator": [
        DecisionTreeClassifier(max_depth=2, random_state=1),
        DecisionTreeClassifier(max_depth=3, random_state=1),
    ],
    "n_estimators": np.arange(50,110,25),
    "learning_rate": np.arange(0.01,0.1,0.05),
```



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Validation

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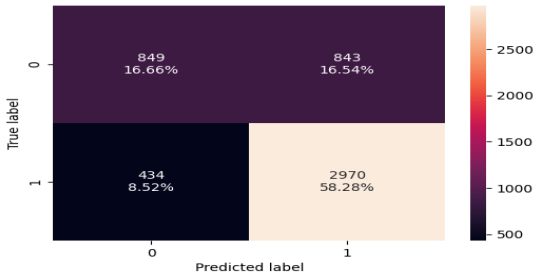
        DecisionTreeClassifier(max_depth=2, random_state=1),
        DecisionTreeClassifier(max_depth=3, random_state=1),
    ],
    "n_estimators": np.arange(50,110,25),
    "learning_rate": np.arange(0.01,0.1,0.05),
```



- **Accuracy: 0.74**
- **Recall: 0.87**
- **Precision: 0.77**
- **F1: 0.82**

The tuned AdaBoost model shows solid validation results with strong recall (87%) and balanced performance (82%) making it a viable model for final testing and comparison.

Hyperparameter Tuning: Gradient Boosting (gbc_tuned)

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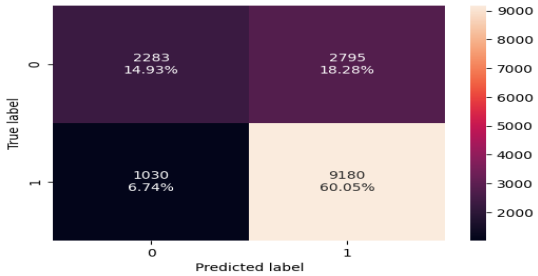
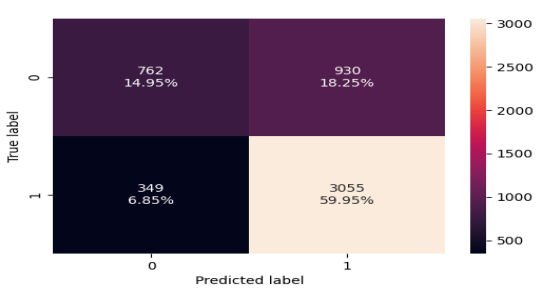
	Tuning	Confusion Matrix	Performance Metrics
Training	`parameters = { "n_estimators": np.arange(50,110,25), "subsample": [0.7,0.9], "max_features": [0.7, 0.8, 0.9, 1], "learning_rate": [0.01,0.1,0.05],`		▪ Accuracy: 0.74 ▪ Recall: 0.89 ▪ Precision: 0.76 ▪ F1: 0.82
Validation	`parameters = { "n_estimators": np.arange(50,110,25), "subsample": [0.7,0.9], "max_features": [0.7, 0.8, 0.9, 1], "learning_rate": [0.01,0.1,0.05],`		▪ Accuracy: 0.74 ▪ Recall: 0.89 ▪ Precision: 0.76 ▪ F1: 0.82

The tuned Gradient Boosting model demonstrates the strongest validation performance, achieving the high recall (89%) and F1-score (82%) suggesting good generalization.

Hyperparameter Tuning: XGBoost Classifier (xgb_tuned)

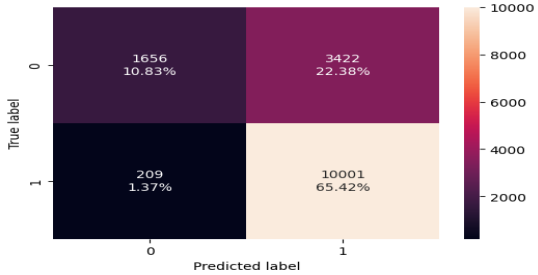
Tuning

Confusion Matrix

Performance Metrics

Training

```
parameters = {
    "n_estimators": np.arange(50,110,25),
    "scale_pos_weight": [1,2,5],
    "subsample": [0.9, 1],
    "learning_rate": [0.01,0.1,0.05],
    "gamma": [1,3]
```



- **Accuracy: 0.76**
- **Recall: 0.97**
- **Precision: 0.74**
- **F1: 0.84**

Validation

```
parameters = {
    "n_estimators": np.arange(50,110,25),
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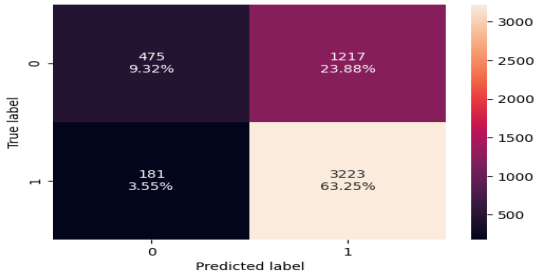
    "subsample": [0.9, 1],
    "learning_rate": [0.01,0.1,0.05],
    "gamma": [1,3]
```



- **Accuracy: 0.72**
- **Recall: 0.94**
- **Precision: 0.72**
- **F1: 0.82**

The tuned XGBoost model achieves the highest recall (94%), making it highly effective at identifying positive cases. This aligns directly with the project objective to minimize false negatives.

Final Model Selection

Training performance comparison:

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.776622	0.749084	0.749804	0.762493
Recall	0.911166	0.876200	0.899119	0.979530
Precision	0.787656	0.776697	0.766597	0.745064
F1	0.844921	0.823454	0.827586	0.846359

Validation performance comparison:

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.749804	0.749411	0.749019	0.725667
Recall	0.890423	0.872503	0.897474	0.946827
Precision	0.770659	0.778914	0.766625	0.725901
F1	0.826223	0.823057	0.826905	0.821775

Best Performing Model

Model Objective

High Recall = Fewer False Negatives

Business Objective is to minimize unexpected denials

High Recall = Fewer False Negatives

Business Objective is to minimize unexpected denials

The tuned XGBoost is the best model because it achieves the highest Recall (94%) , aligning directly with the business objective of minimizing high-risk visa approvals while maintaining a strong F1-score balance.

Best Model: XGBoost Classifier (xgb_tuned)

Training Data Results

	Accuracy	Recall	Precision	F1
0	0.713108	0.947121	0.71549	0.815171

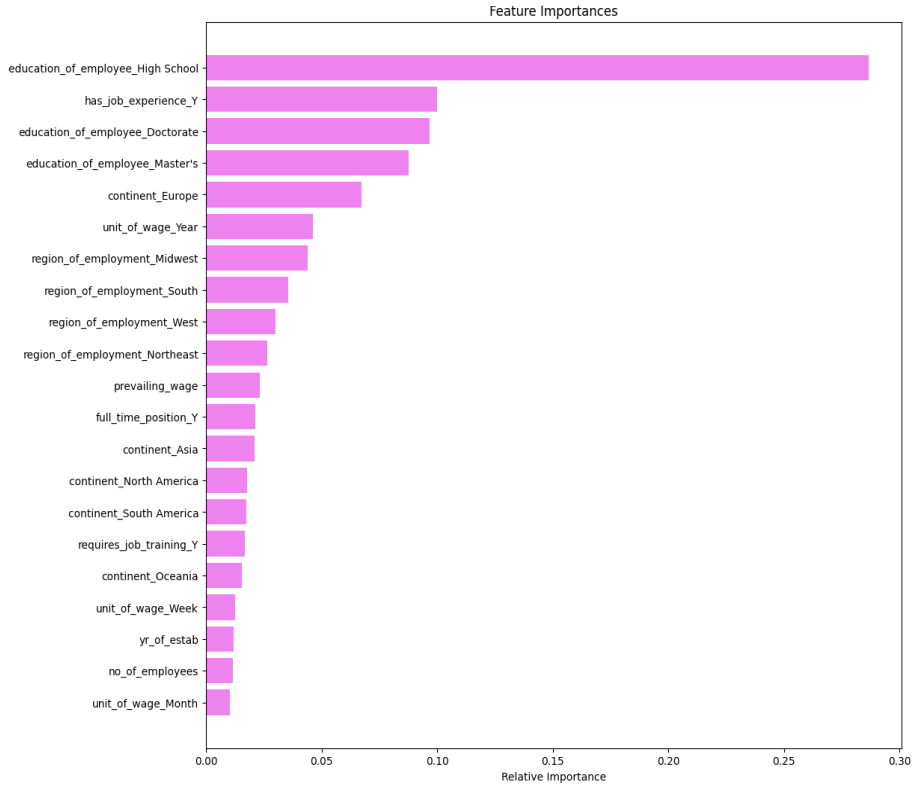
The tuned XGBoost model was selected due to its strong performance, achieving 94.7% recall and an F1 score of 81.5%.
It is optimized to prioritize identifying high-risk visa applications, minimizing missed denials.

Most relevant features include:
- Education
- Job Experience
- Wage Level
- Geographic Region

Tuned Parameters

Optimal Parameters Include:
- ✓ n_estimator = 100
- ✓ learning_rate = 0.1
- ✓ gamma = 1
- ✓ subsample = 1 <default>

Feature Importance



Conclusions & Recommendations

The EasyVisa predictive model successfully addresses the core challenge of identifying high-risk visa applications before submission. Key factors such as education, job experience, employment type, wage level, region, and employer characteristics significantly influence certification outcomes. Among the models tested, the tuned XGBoost model delivered the strongest performance, achieving 94.7% recall and an 81.5% F1-score, effectively minimizing high-risk approvals while maintaining balanced performance.

Recommendations:

- **Model Deployment**: Integrate the tuned XGBoost model into the pre-submission workflow.
- **Business Integration**: Focus on wage compliance, job structuring, and qualifying positioning.
- **Performance Monitoring**: Continuously track key performance metrics—including recall, precision, and overall model effectiveness—and incorporate periodic retraining into the model lifecycle to maintain accuracy, manage data drift and ensure sustained performance.
- **Data Enhancement**: Incorporate relevant external data and refine the features to improve predictive accuracy.
- **Approval process:** Implement structure to risk categories (High, Medium, Low) to prioritize the approval process.

APPENDIX

Data Background and Contents

Statistical Summary

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.0	5667.043210	22877.928848	-26.0000	1022.00	2109.00	3504.0000	602069.00
yr_of_estab	25480.0	1979.409929	42.366929	1800.0000	1976.00	1997.00	2005.0000	2016.00
prevailing_wage	25480.0	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27

The dataset contains 25,480 observations with key numerical variables including number of employees, year of establishment, and prevailing wage.

- The **number of employees** variable shows strong right skewness with extreme outliers and at least one invalid negative value.
- The **year of establishment** appears reasonable with most companies founded after the 1970s.
- The **prevailing wage** variable shows moderate variability with a wide range and a few unusually low values.

Overall, the data is usable but requires minor cleaning and outlier handling before model development.

Great Learning

Happy Learning !